Double Integrals

ANSWER KEY

Section 1: Fubini's Theorem over Rectangles

1 Evaluate $\int_0^2 \int_0^3 2xy\,dy\,dx$.

a. 18 ✓

b. 9

c. 36

d. 12

Inner integral: $\int_0^3 2xy\,dy = [xy^2]_0^3 = 9x$. Outer integral: $\int_0^2 9x\,dx = [\frac{9}{2}x^2]_0^2 = \frac{9}{2}(4) = 18$.

2 What does Fubini's Theorem state for a continuous function $f(x, y)$ over a rectangle $R = [a, b] \times [c, d]$?

a. **The iterated integrals in either order are equal:** $\int_a^b \int_c^d f(x, y)\,dy\,dx = \int_c^d \int_a^b f(x, y)\,dx\,dy$ ✓

b. The double integral is always equal to $f(b, d) - f(a, c)$

c. The integral is zero if f is odd in x

d. The integral must be evaluated with x first

Fubini's Theorem guarantees that for continuous functions over rectangles, the order of integration does not affect the value of the double integral.

3 Evaluate $\iint_R (x + y)\,dA$ over $R = [0, 1] \times [0, 2]$.

a. 3 ✓

b. 2

c. 4

d. 1

$\int_0^1 \int_0^2 (x + y)\,dy\,dx = \int_0^1 [xy + \frac{y^2}{2}]_0^2 dx = \int_0^1 (2x + 2)\,dx = [x^2 + 2x]_0^1 = 1 + 2 = 3$.

Section 2: Double Integrals over General Regions

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4 Set up the double integral of $f(x, y)$ over the triangular region bounded by $y = 0$, $y = 2x$, and $x = 3$ as a Type I region.

a. $\int_0^3 \int_0^{2x} f(x, y)dydx$ ✓

b. $\int_0^6 \int_0^3 f(x, y)dxdy$

c. $\int_0^{2x} \int_0^3 f(x, y)dxdy$

d. $\int_0^3 \int_{2x}^6 f(x, y)dydx$

For a Type I region, x varies between constants $0 \le x \le 3$, and for each x, y goes from bottom boundary $y = 0$ to top boundary $y = 2x$: $\int_0^3 \int_0^{2x} f(x, y)dydx$.

5 Evaluate $\int_0^1 \int_0^x 6xydydx$.

a. $\frac{3}{4}$ ✓

b. 1

c. $\frac{1}{2}$

d. $\frac{3}{2}$

Inner: $\int_0^x 6xydy = [3xy^2]_0^x = 3x^3$. Outer: $\int_0^1 3x^3dx = [\frac{3}{4}x^4]_0^1 = \frac{3}{4}$.

6 What geometric quantity does $\iint_D 1dA$ compute?

a. **The area of the planar region** D ✓

b. The perimeter of region D

c. The volume under the paraboloid

d. The centroid

Integrating the constant height 1 over a 2D domain D calculates the area of D: $\text{Area}(D) = \iint_D dA$.

Section 3: Changing the Order of Integration

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7 To evaluate $\int_0^1 \int_y^1 e^{x^2} dx dy$, which cannot be integrated directly, reverse the order of integration.

a. $\int_0^1 \int_0^x e^{x^2} dy dx$ ✓

b. $\int_0^1 \int_x^1 e^{x^2} dy dx$

c. $\int_0^1 \int_0^1 e^{x^2} dx dy$

d. $\int_y^1 \int_0^1 e^{x^2} dy dx$

Region: $0 \le y \le 1$ and $y \le x \le 1$. Inverting: $0 \le x \le 1$ and $0 \le y \le x$, yielding $\int_0^1 \int_0^x e^{x^2} dy dx$.

8 Evaluate the reversed integral $\int_0^1 \int_0^x e^{x^2} dy dx$.

a. $\frac{e-1}{2}$ ✓

b. $e - 1$

c. $\frac{e}{2}$

d. 1

Inner: $\int_0^x e^{x^2} dy = x e^{x^2}$. Outer: $\int_0^1 x e^{x^2} dx = [\frac{1}{2} e^{x^2}]_0^1 = \frac{1}{2}(e^1 - e^0) = \frac{e-1}{2}$.

9 Reverse the order of integration for $\int_0^4 \int_{\sqrt{x}}^2 f(x, y) dy dx$.

a. $\int_0^2 \int_0^{y^2} f(x, y) dx dy$ ✓

b. $\int_0^2 \int_{y^2}^4 f(x, y) dx dy$

c. $\int_0^4 \int_0^{y^2} f(x, y) dx dy$

d. $\int_{\sqrt{x}}^2 \int_0^4 f(x, y) dx dy$

Region: $0 \le x \le 4$, $\sqrt{x} \le y \le 2$. The curve is $y = \sqrt{x} \Rightarrow x = y^2$. For $y \in [0, 2]$, x ranges from 0 to y^2: $\int_0^2 \int_0^{y^2} f(x, y) dx dy$.

Section 4: Double Integrals in Polar Coordinates

10 What is the differential area element dA in polar coordinates?

a. $dA = r dr d\theta$ ✓

b. $dA = dr d\theta$

c. $dA = r^2 dr d\theta$

d. $dA = \frac{1}{r} dr d\theta$

The Jacobian of the transformation $x = r\cos\theta, y = r\sin\theta$ is r, giving area element $dA = r dr d\theta$.

11 Evaluate $\iint_D e^{-(x^2+y^2)} dA$ where D is the disk $x^2 + y^2 \le 4$.

a. $\pi(1 - e^{-4})$ ✓

b. $2\pi(1 - e^{-4})$

c. πe^{-4}

d. $\frac{\pi}{2}(1 - e^{-4})$

In polar: $\int_0^{2\pi} \int_0^2 e^{-r^2} r \, dr d\theta = 2\pi[-\frac{1}{2}e^{-r^2}]_0^2 = 2\pi(-\frac{1}{2}e^{-4} + \frac{1}{2}) = \pi(1 - e^{-4})$.

12 Convert $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx$ to polar coordinates.

a. $\int_0^{\pi/2} \int_0^1 r^3 dr d\theta$ ✓

b. $\int_0^{2\pi} \int_0^1 r^2 dr d\theta$

c. $\int_0^{\pi/2} \int_0^1 r^2 dr d\theta$

d. $\int_0^1 \int_0^1 r^3 dr d\theta$

The region is the quarter disk in Quadrant I ($0 \le \theta < \pi/2$, $0 \le r \le 1$). The integrand is $x^2 + y^2 = r^2$, and $dA = r dr d\theta$, giving $\int_0^{\pi/2} \int_0^1 r^2 \cdot r \, dr d\theta = \int_0^{\pi/2} \int_0^1 r^3 dr d\theta$.

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